Online Constrained Control of Storage Systems via Simplex Disturbance-Action Policies

Kamiar Asgari and Michael J. Neely

***Abstract*—We study online control of a scalar system under adversarial disturbances, convex cost functions, and state-dependent action constraints. At each time, the controller must choose a feasible action before observing the corresponding disturbance and cost. We introduce *Simplex Disturbance-Action Control* (SDAC), a simplex-constrained disturbance-action policy class that preserves per-step feasibility by construction. An online controller updates SDAC parameters via online mirror descent (OMD) on the sub-probability simplex, with a feasibility projection that enforces the state-dependent action constraint. We prove that regret against the best fixed SDAC policy in hindsight is sublinear in the horizon and logarithmic in the policy dimension, implying no-regret relative to feasible linear state-feedback policies. The setting is motivated by energy-harvesting battery management and similar storage systems.**

***Index Terms*—Online control, disturbance-action policies, storage systems, energy harvesting, regret minimization, state constraints.**

I. Introduction

A. Motivation and Problem Setting

MANY dynamic resource systems can be modeled by a storage state that evolves over time and constrains the control action at every time. The controller must choose an action subject to state-dependent constraints before observing the corresponding disturbance and cost.

This creates a constrained discrete-time control problem with state–control coupling: aggressive early control actions can harm later performance, while overly conservative actions can increase per-step cost. Our objective is to design a policy that remains feasible at every time and competes with the best fixed policy in a rich policy class—one expressive enough to capture many strong policies.

B. Model and Information Structure

We study a discrete-time storage system with state $x_t \geq 0$, where x_t models the storage level at the beginning of time t. At each time t, the controller chooses a control action u_t under the *state constraints*

$$0 \leq u_t \leq \alpha x_t,$$

where $\alpha \in (0, 1)$ is a fixed known state-retention factor (so a fraction $1-\alpha$ of the state is lost each period). After the action is chosen, an adversarial disturbance w_t arrives, and the state evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t.$$

Unlike stochastic application models that rely on distributional assumptions, our formulation is adversarial. Disturbances w_t and costs $c_t(x, u)$ may vary arbitrarily over time, and the controller chooses u_t before observing either at time t. The goal is low cumulative cost relative to the best fixed policy in a later-defined policy class, while both our algorithm and the benchmark policy maintain feasibility throughout.

C. Application Domains

This abstraction captures a broad set of domains:

- **Energy Harvesting and Battery Management:** x_t is battery state of charge (retention factor α after self-discharge), w_t is harvested energy, and u_t is discharge or consumption bounded by available energy αx_t.
- **Queueing Networks and Buffer Management:** x_t is queue/buffer occupancy (retention factor α after abandonment/leakage), w_t is packet arrivals, and u_t is service bounded by αx_t.
- **Inventory Control:** x_t is on-hand inventory (retention factor α after spoilage/deterioration), w_t is supply or replenishment, and u_t is sales bounded by αx_t.
- **Investments and Bank Accounts:** x_t is liquid capital/balance (retention factor α after fees), w_t is incoming cash flow, and u_t is capital deployment or withdrawal bounded by αx_t.
- **Water Reservoir and Dam Management:** x_t is stored water (retention factor α after evaporation/seepage), w_t is inflow, and u_t is controlled release bounded by αx_t.

a) Energy Harvesting and Battery Management: A closely related work is [1], which studies an online energy-harvesting allocation problem with dynamics similar to ours. That work benchmarks against the best sequence of decisions in hindsight, but it differs in several important ways: [1] uses competitive ratio rather than regret, adopts an action-only cost model rather than general convex state-action costs, which means in particular that it cannot optimize a function of the battery size, and it has a less adversarial information pattern because the algorithm observes the current cost before selecting the action, whereas in our model the control action is chosen before the cost is revealed.

Prior work on energy harvesting includes [2], which studies online allocation with stochastic costs, and [3], which improves over [2] by achieving $O(\sqrt{T})$ regret with finite

K. Asgari and M. J. Neely are with the Department of Electrical and Computer Engineering, University of Southern California, Los Angeles, CA 90089 USA (e-mail: kamiaras@usc.edu; mjneely@usc.edu).

Corresponding author: K. Asgari.

battery and adversarial costs. Both works consider action-only costs (not state-dependent), i.i.d. disturbances (not adversarial), and benchmark against fixed allocation amounts rather than fixed policies. In contrast, our framework jointly optimizes over state-action costs, adversarial disturbances, and stronger benchmarks, requiring new algorithmic techniques.

b) Queueing and Network Optimization: An alternative, though mathematically related, perspective comes from queueing control in computer networks and telecommunications. In that framework, one tracks a queue with stochastic arrivals and makes real-time service decisions, typically aiming to minimize long-run average service cost while keeping queue lengths stable. This literature is extensive (see, e.g., [4]–[7]), but its setting differs from ours in several important respects. Classical queueing work usually assumes stochastic arrivals and focuses on simple, known cost functions that depend only on the service rate, while queue length is kept small. These methods achieve trade-offs between queue size and time-average service cost. By contrast, our model permits fully adversarial sequences of disturbances and cost functions and allows general convex costs that depend on both the queue length (state) and the service rate (control action). Thus our approach is more general in cost structure and exogenous-disturbance model. The queueing literature provides useful motivation and techniques for state-coupled dynamics, but does not directly address the online optimization and regret-minimization objectives we study.

Another related work is [8], which proposes and analyzes a fixed controller that is a linear function of past arrivals. That controller resembles our family of SDAC policies because it uses nonnegative filter weights that sum to one (summing to at most one for SDAC). However, the controller in [8] is not updated online and is chosen to achieve bounded service delay; by contrast, our framework focuses on online optimization of a general cost and does not treat delay as a constraint.

c) Inventory Management: A related work is [9], which studies a distinct model in which demand (equivalent to allocation in our setup) is chosen adversarially, while the controller chooses the supply (equivalent to disturbances in our model). The objective is to minimize time-average purchasing cost while always meeting demand, and the performance metric is competitive ratio rather than regret. Although structurally similar to our setting, the control variable differs: we choose the control action from the feasible state bound, whereas [9] chooses how much to procure. Another key difference is the cost model: our framework allows general convex state-action costs, while [9] uses a linear cost depending only on the control action.

More closely related to ours is [10], which considers an inventory-control formulation with potentially nonlinear and nondifferentiable state-transition functions, and studies base-stock policies. While that model is more general in scope, its decision structure differs from ours: as in [9], the controller chooses supply (inflow) and then observes demand (allocation) through the transition, whereas our controller chooses the control action and observes incoming supply (disturbances). Methodologically, our setup is intentionally narrower and tailored to SDAC-based adversarial regret analysis: feasibility constraints are enforced by policy design, yielding a simpler linear state recursion, lower computational overhead, and cleaner analysis than approaches operating through nonlinear nondifferentiable dynamics such as [10]. The two frameworks are structurally related but target different problem structures, control variables, and policy classes.

d) Water Reservoir and Dam Management: Classical dam theory studies storage dynamics with random inflows and controlled releases (see, for example, [11], [12]). Relative to our adversarial online-control setting, this literature typically assumes stochastic inputs and emphasizes stationary performance criteria under simple, known cost structures. Here, it serves mainly as historical context for state-coupled control under state constraints.

e) Online control: A recent line of work in online learning studies online control, where a linear dynamical system is operated under adversarial disturbances and costs. The objective is low time-average cost relative to the best policy in hindsight, rather than classical stabilization or worst-case guarantees. Foundational results for linear systems and adversarial settings include [13]–[16]; see [17] for a modern overview.

A central idea in this literature is *Disturbance-Action Control* (DAC), a policy parameterization in which control actions are expressed as affine or linear functions of the current state and a finite history of past disturbances. This representation yields tractable surrogates for policy optimization and enables efficient no-regret updates in adversarial environments. Extensions include partial observability [18] and bandit feedback [15]. SDAC adapts this template to state-constrained resource systems, where feasibility is coupled over time through the dynamics.

Online control provides a strong foundation for our problem class: it accommodates arbitrary time-varying disturbances and unknown general convex cost functions, and the resulting algorithms come with robust guarantees and tractability. However, for state-coupled storage dynamics, unconstrained DAC does not enforce per-step feasibility by construction. Our framework fills this gap by replacing DAC with SDAC, a simplex-constrained DAC-style family that preserves the feasibility defined by (2) while retaining the desirable characteristics.

D. Contributions

The contributions are organized in two layers:

- **Modeling and policy-class contribution.** We formulate adversarial online control of a storage system with retention-coupled state dynamics, state-dependent feasibility ($0 \leq u_t \leq \alpha x_t$), and state-dependent cost functions, and we introduce *Simplex Disturbance-Action Control* (SDAC), a DAC-inspired simplex-constrained policy class to compete with. The SDAC *enforces feasibility* by construction while preserving DAC-style expressiveness.
- **Algorithmic and analytical contribution.** We develop an OMD controller with feasibility projection over SDAC parameters that *enforces* the state-dependent constraint on the online control action, and we analyze regret against the best fixed SDAC policy in hindsight, proving a bound

that is sublinear in the horizon and logarithmic in the policy dimension.

E. Paper Organization

Section III formalizes the adversarial storage-control model, assumptions, and regret benchmark. Section IV introduces the SDAC policy class and summarizes its structural properties. Section V presents the online controller, Section VI develops regret guarantees, and Section VII concludes. Deferred proofs appear in the Appendix.

II. Notation

We write $[z]_+ \triangleq \max\{z, 0\}$. For $v \in \mathbb{R}^n$, the i-th coordinate is $v^{(i)}$, and $\mathbf{1}$ denotes the all-ones vector (dimension clear from context). The Euclidean inner product is $\langle \cdot, \cdot \rangle$. Time indices may be nonpositive, with corresponding quantities taken to be zero (e.g. $w_\tau \equiv 0$ for $\tau \leq 0$). We write $\mathcal{A}$ for the online algorithm and π (or $\pi(M)$) for a fixed policy.

III. Model and Problem Setup

We consider a discrete-time process over times $t \in \{1, 2, \ldots, T\}$. The system state is denoted by $x_t \in \mathbb{R}_{\geq 0}$, where x_t represents the resource level at the beginning of time t. The initial state is given as

$$x_1 = 0. \tag{1}$$

Fix a state-retention factor $\alpha \in (0, 1)$. At each time, the controller chooses a control action $u_t \in \mathbb{R}_{\geq 0}$ subject to the state constraint:

$$0 \leq u_t \leq \alpha x_t. \tag{2}$$

The system then receives a disturbance $w_t \in [0, 1]$ and evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t. \tag{3}$$

After the control action is selected, a convex cost function

$$c_t : \mathbb{R}^2_{\geq 0} \to \mathbb{R}$$

is revealed, and the incurred cost is $c_t(x_t, u_t)$. We assume each c_t is convex in (x, u) and satisfies a Lipschitz condition on the relevant domain. There exist constants $L_x, L_u > 0$ such that for all admissible (x, u) and (x', u'),

$$|c_t(x, u) - c_t(x', u')| \leq L_x|x - x'| + L_u|u - u'|. \tag{4}$$

The information pattern is adversarial and non-anticipatory. At time t, the controller chooses u_t before observing w_t and c_t. Hence both the disturbance sequence $\{w_t\}_{t=1}^T$ and the sequence of cost functions $\{c_t\}_{t=1}^T$ may be arbitrary.

a) Assumptions:

1) **Retention factor:** $\alpha \in (0, 1)$ is fixed and known (a fraction $1 - \alpha$ of the state is lost each period).
2) **Disturbance bounds:** $w_t \in [0, 1]$ for all t.
3) **Adversarial timing:** at time t, u_t is chosen before observing w_t and c_t.
4) **Cost regularity:** each c_t is convex in (x, u) and obeys (4).
5) **Feasible policy class:** both the online policy to be developed and the benchmark described below are required to satisfy (2) under dynamics (3).

For an online algorithm $\mathcal{A}$ producing trajectory $\{(x_t^{\mathcal{A}}, u_t^{\mathcal{A}})\}_{t=1}^T$, define cumulative cost

$$J_T(\mathcal{A}) \triangleq \sum_{t=1}^{T} c_t\big(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}\big). \tag{5}$$

For a fixed positive integer H, let π range over the policy class SDAC_H (defined in Section IV). For each $\pi \in \mathrm{SDAC}_H$, let $\{(x_t^\pi, u_t^\pi)\}_{t=1}^T$ be the induced feasible trajectory under the same disturbance sequence. Regret against the best fixed SDAC policy is defined by

$$\mathrm{Reg}_{\mathrm{SDAC}}(T) \triangleq \sum_{t=1}^{T} c_t\big(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}\big) - \min_{\pi \in \mathrm{SDAC}_H} \sum_{t=1}^{T} c_t\big(x_t^{\pi}, u_t^{\pi}\big). \tag{6}$$

While the online controller may update its SDAC parameters over time, the benchmark in (6) is the best single fixed SDAC policy chosen in hindsight. The objective is to design an online policy that preserves feasibility in (2) for all t and achieves sublinear regret as defined in (6).

IV. Simplex Disturbance-Action Control (SDAC)

A. From Disturbance-Action Control to SDAC

Disturbance-Action Control (DAC) parameterizes the control action as a linear function of a finite window of past disturbances. This representation is central in adversarial online control because the cost is convex in term of the parameters, yields robust regret guarantees, and enables efficient online convex-optimization updates. In our setting, a representative DAC form is

$$u_t = \sum_{i=1}^{H} M^{(i)} w_{t-i}, \tag{7}$$

where $H \in \mathbb{N}$ is the memory horizon and $M \in \mathbb{R}^H$ is a linear filter. We allow nonpositive indices in such sums, with the convention $w_\tau \equiv 0$ for $\tau \leq 0$.

For our constrained model, unconstrained DAC is insufficient. In particular, the control action can violate (2), either by having $u_t > \alpha x_t$ or $u_t < 0$. We therefore introduce *Simplex Disturbance-Action Control* (SDAC), which replaces unconstrained DAC filters by a simplex-constrained family tailored to this dynamics. SDAC is designed to keep the DAC advantages above (adversarial robustness, regret-based comparison, and tractable online updates) while adding feasibility by construction.

B. SDAC Definition

Fix memory horizon $H \in \mathbb{N}$. Let

$$\Delta_H \triangleq \left\{ M \in \mathbb{R}^H_{\geq 0} : \sum_{i=1}^{H} M^{(i)} \leq 1 \right\} \tag{8}$$

be the sub-probability simplex. For any $M \in \Delta_H$, define the policy $\pi(M)$ by

$$u_t^{\pi(M)} \triangleq \sum_{i=1}^{H} M^{(i)} \alpha^i w_{t-i}, \qquad w_\tau \equiv 0 \text{ for } \tau \leq 0. \tag{9}$$

For each time $t \in \{1, 2, \dots, T\}$, let

$$\widetilde{w}_t \triangleq \left(\alpha w_{t-1}, \alpha^2 w_{t-2}, \dots, \alpha^H w_{t-H}\right)^\top \in \mathbb{R}^H, \tag{10}$$

with the same convention $w_\tau \equiv 0$ for $\tau \leq 0$, so that (9) can be written as

$$u_t^{\pi(M)} = \langle M, \widetilde{w}_t \rangle. \tag{11}$$

The induced state trajectory satisfies

$$x_{t+1}^{\pi(M)} = \alpha x_t^{\pi(M)} - u_t^{\pi(M)} + w_t, \qquad x_1^{\pi(M)} = 0. \tag{12}$$

The weights α^i align the disturbance filter with the retention factor α in (3).

We now establish the two structural properties that make SDAC useful: the induced control action always satisfies the feasibility constraint (2), and the induced cost is convex and Lipschitz in M, so that regret minimization against the best fixed SDAC policy is an instance of online convex optimization over the simplex. Both properties follow from an explicit decomposition of the state trajectory that isolates its dependence on M.

Recall that we set $w_\tau \equiv 0$ for $\tau \leq 0$, which is used in the base-case arguments below.

Lemma 1 (State decomposition under SDAC)**.** *Fix $H \in \mathbb{N}$ and $M \in \Delta_H$, and write $W \triangleq \sum_{i=1}^{H} M^{(i)} \leq 1$. Let $\pi(M)$ be defined by* (9) *and* (12)*. Then for all $t \geq 2$,*

$$x_t^{\pi(M)} = \sum_{i=1}^{H} M^{(i)} \psi_t^{(i)} + \big(1 - W\big)\rho_t = \langle M, \psi_t \rangle + (1 - W)\rho_t, \tag{13}$$

where

$$\psi_t^{(i)} \triangleq \sum_{j=1}^{i} \alpha^{j-1} w_{t-j} \quad (i = 1, \dots, H),$$

and

$$\rho_t \triangleq \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j}.$$

Equivalently, ρ_t is the state $x_t^{\pi(0)}$ produced by the null policy $M = 0$ (release nothing, $x_{\tau+1} = \alpha x_\tau + w_\tau$), and the coefficients satisfy $0 \leq \psi_t^{(i)} \leq \rho_t \leq 1/(1-\alpha)$. In particular, $M \mapsto x_t^{\pi(M)}$ is affine.

Proof. Isolating the $j = 1$ term in the definition of ρ_{t+1} and reindexing gives the one-step identity $\rho_{t+1} = \alpha\rho_t + w_t$ (with $\rho_1 = 0$). We prove (13) by induction on t.

Base case $t = 2$. Since $x_1^{\pi(M)} = 0$ and $w_{1-i} = 0$ for all $i \geq 1$, the policy (9) gives $u_1^{\pi(M)} = \sum_{i=1}^{H} M^{(i)} \alpha^i w_{1-i} = 0$, and the update (12) gives $x_2^{\pi(M)} = \alpha x_1^{\pi(M)} - u_1^{\pi(M)} + w_1 = w_1$. On the other hand $\psi_2^{(i)} = w_1$ for every i and $\rho_2 = w_1$, so $\langle M, \psi_2 \rangle + (1 - W)\rho_2 = W w_1 + (1 - W) w_1 = w_1$, matching (13).

Inductive step. Assume (13) holds at time $t \geq 2$. Using (9), (12), and the induction hypothesis,

$$\begin{aligned} x_{t+1}^{\pi(M)} &= \alpha x_t^{\pi(M)} - \sum_{i=1}^{H} M^{(i)} \alpha^i w_{t-i} + w_t \\ &= \Big(\alpha\langle M, \psi_t \rangle - \sum_{i=1}^{H} M^{(i)} \alpha^i w_{t-i}\Big) \\ &\quad + \alpha(1 - W)\rho_t + w_t. \end{aligned}$$

Reindexing the inner summation ($j \mapsto j + 1$) yields

$$\begin{aligned} &\alpha\langle M, \psi_t \rangle - \sum_{i=1}^{H} M^{(i)} \alpha^i w_{t-i} \\ &\quad = \sum_{i=1}^{H} M^{(i)} \left(\sum_{j=1}^{i-1} \alpha^j w_{t-j} \right) \\ &\quad = \sum_{i=1}^{H} M^{(i)} \big(\psi_{t+1}^{(i)} - w_t\big) = \langle M, \psi_{t+1} \rangle - W w_t, \end{aligned}$$

where the second equality uses $\psi_{t+1}^{(i)} = w_t + \sum_{j=1}^{i-1} \alpha^j w_{t-j}$. Substituting this together with $\rho_{t+1} = \alpha\rho_t + w_t$,

$$\begin{aligned} x_{t+1}^{\pi(M)} &= \langle M, \psi_{t+1} \rangle - W w_t + \alpha(1 - W)\rho_t + w_t \\ &= \langle M, \psi_{t+1} \rangle + (1 - W)\big(w_t + \alpha\rho_t\big) \\ &= \langle M, \psi_{t+1} \rangle + (1 - W)\rho_{t+1}, \end{aligned}$$

which is (13) at time $t + 1$.

Bounds. Each $\psi_t^{(i)}$ is the truncation to $j \leq i$ of the nonnegative series defining ρ_t, so $0 \leq \psi_t^{(i)} \leq \rho_t$; and since $w_\tau \in [0, 1]$, the geometric series gives $\rho_t = \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j} \leq \sum_{j=1}^{\infty} \alpha^{j-1} = 1/(1-\alpha)$. Affineness in M is immediate, as ψ_t and ρ_t do not depend on M. $\square$

The first consequence of this decomposition is that feasibility is preserved within the class.

Lemma 2 (Feasibility of SDAC policies)**.** *For every fixed $M \in \Delta_H$, the trajectory induced by* (9)–(12) *satisfies*

$$0 \leq u_t^{\pi(M)} \leq \alpha x_t^{\pi(M)} \quad \textit{for all } t \geq 1.$$

Proof. Nonnegativity of $u_t^{\pi(M)}$ follows from (9), because $M^{(i)} \geq 0$, $\alpha^i \geq 0$, and $w_{t-i} \geq 0$ for every i.

For the upper bound at $t = 1$, we have $x_1^{\pi(M)} = 0$ and $w_{1-i} = 0$ for all $i \geq 1$, so $u_1^{\pi(M)} = 0 = \alpha x_1^{\pi(M)}$.

Now fix $t \geq 2$. Lemma 1 gives $x_t^{\pi(M)} = \langle M, \psi_t \rangle + (1 - W)\rho_t$ with $W = \sum_{i=1}^{H} M^{(i)} \leq 1$. Since $(1 - W)\rho_t \geq 0$,

$$\begin{aligned} \alpha x_t^{\pi(M)} &\geq \alpha\langle M, \psi_t \rangle = \alpha \sum_{i=1}^{H} M^{(i)} \left(\sum_{j=1}^{i} \alpha^{j-1} w_{t-j} \right) \\ &= \sum_{i=1}^{H} M^{(i)} \left(\sum_{j=1}^{i} \alpha^j w_{t-j} \right). \end{aligned}$$

Collecting the coefficient of each w_{t-i} and using that all parameters are nonnegative, for every i we have $\sum_{j=i}^{H} M^{(j)} \geq M^{(i)}$, so

$$\begin{aligned} \alpha x_t^{\pi(M)} &\geq \sum_{i=1}^{H} \left(\sum_{j=i}^{H} M^{(j)} \right) \alpha^i w_{t-i} \\ &\geq \sum_{i=1}^{H} M^{(i)} \alpha^i w_{t-i} = u_t^{\pi(M)}, \end{aligned}$$

since $w_{t-i} \geq 0$. □

The second consequence is that SDAC is compatible with online convex optimization: for fixed disturbance sequences, $u_t^{\pi(M)}$ is linear in M by (11), and $x_t^{\pi(M)}$ is affine in M by Lemma 1. Since each c_t is convex in (x, u), this already shows the induced cost is convex in M; the next lemma also quantifies its Lipschitz constant, which we use later to tune the online algorithm's step size.

Lemma 3 (Convexity and Lipschitz continuity of surrogate cost)**.** *Define*

$$\ell_t(M) \triangleq c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big), \qquad M \in \Delta_H.$$

Then ℓ_t is convex on Δ_H, and for all $M, M' \in \Delta_H$,

$$|\ell_t(M) - \ell_t(M')| \leq \left(\frac{L_x}{1-\alpha} + \alpha L_u \right) \|M - M'\|_1.$$

Proof. By Lemma 1 and (11), with $W = \sum_{i=1}^{H} M^{(i)}$,

$$\ell_t(M) = c_t\big(\langle M, \psi_t \rangle + (1 - W)\rho_t,\ \langle M, \widetilde{w}_t \rangle\big).$$

The map $M \mapsto \big(x_t^{\pi(M)}, u_t^{\pi(M)}\big)$ is affine (Lemma 1 and (11)) and c_t is convex, so ℓ_t is convex.

For the Lipschitz estimate, the decomposition gives, for all $M, M' \in \Delta_H$,

$$\begin{aligned} x_t^{\pi(M)} - x_t^{\pi(M')} &= \langle M - M', \psi_t \rangle - (W - W')\rho_t \\ &= \langle M - M',\, \psi_t - \rho_t \mathbf{1} \rangle, \end{aligned}$$

using $W - W' = \langle M - M', \mathbf{1} \rangle$, while $u_t^{\pi(M)} - u_t^{\pi(M')} = \langle M - M', \widetilde{w}_t \rangle$. Hence, by (4) and Hölder's inequality,

$$\begin{aligned} |\ell_t(M) - \ell_t(M')| &\leq L_x \big| \langle M - M',\, \psi_t - \rho_t \mathbf{1} \rangle \big| \\ &\quad + L_u \big| \langle M - M', \widetilde{w}_t \rangle \big| \\ &\leq \big(L_x \, \|\psi_t - \rho_t \mathbf{1}\|_\infty + L_u \, \|\widetilde{w}_t\|_\infty \big) \\ &\quad \cdot \|M - M'\|_1. \end{aligned}$$

Since $0 \leq \psi_t^{(i)} \leq \rho_t \leq 1/(1-\alpha)$ (Lemma 1), we have $\|\psi_t - \rho_t \mathbf{1}\|_\infty = \max_i \big(\rho_t - \psi_t^{(i)}\big) \leq \rho_t \leq 1/(1-\alpha)$, and $\|\widetilde{w}_t\|_\infty = \max_i \alpha^i w_{t-i} \leq \alpha$ because $w_\tau \in [0, 1]$ and $i \geq 1$. This yields the claim, and one may take the surrogate Lipschitz constant

$$L := \alpha L_u + \frac{L_x}{1-\alpha},$$

as used in the main text. □

Regret minimization against the best fixed SDAC policy thus becomes online convex optimization over the simplex.

C. Expressiveness: Approximating Linear State-Feedback Policies

A standard policy class in online control is the family of linear state-feedback policies $u_t = -K x_t$, and a foundational property of Disturbance-Action Control is that the DAC class *approximates* any strongly stable linear policy, with approximation error decaying geometrically in the memory horizon H [13], [17].

We establish the analogue of this guarantee for SDAC. In our scalar, state-constrained model the feasible linear state-feedback policies are exactly the proportional (constant-fraction) control actions $u_t = q\, x_t$ with gain $q \in [0, \alpha]$: since $x_t \geq 0$, the feasibility constraint $0 \leq u_t \leq \alpha x_t$ in (2) holds if and only if $q \in [0, \alpha]$. The next lemma shows that every such policy with $q \in (0, \alpha)$ is approximated, *within the feasible class* SDAC_H, by explicit parameters, with cost approximation error that decays geometrically in H.

Lemma 4 (SDAC approximates linear state-feedback policies)**.** *Fix $q \in (0, \alpha)$, set $\beta \triangleq \alpha - q$, and let π_q^{lin} denote the feasible linear policy*

$$u_t^{\mathrm{lin}} = q\, x_t^{\mathrm{lin}}, \qquad x_{t+1}^{\mathrm{lin}} = \beta\, x_t^{\mathrm{lin}} + w_t, \qquad t \geq 1, \qquad x_1^{\mathrm{lin}} = 0.$$

Define $M_q^\star \in \mathbb{R}^H$ by

$$M_q^{\star(i)} \triangleq \frac{q}{\alpha} \Big(\frac{\beta}{\alpha} \Big)^{i-1}, \qquad i = 1, \ldots, H.$$

Then $M_q^\star \in \Delta_H$, and if each c_t satisfies (4)*, then for every $\{w_t\} \subset [0, 1]$,*

$$\begin{aligned} &\Big| \sum_{t=1}^{T} c_t\big(x_t^{\pi(M_q^\star)}, u_t^{\pi(M_q^\star)}\big) - \sum_{t=1}^{T} c_t\big(x_t^{\mathrm{lin}}, u_t^{\mathrm{lin}}\big) \Big| \\ &\qquad \leq \Big(L_u + \frac{L_x}{1-\alpha} \Big) \frac{q\, \beta^H}{1-\beta}\, T. \end{aligned} \tag{14}$$

In particular, the cost error decays geometrically in H, so SDAC_H approximates π_q^{lin} arbitrarily well as $H \to \infty$. The proof is deferred to Appendix A.

This expressiveness property has a direct algorithmic consequence. Because the regret bound of Theorem 1 depends on the memory horizon only through $\sqrt{\log(H+1)}$, the window H can be enlarged cheaply; combining Lemma 4 with that bound then shows the online controller competes with the best fixed linear policy. We make this precise in Corollary 1, mirroring the role of DAC-to-linear approximation in online control [13].

V. Online Algorithm

We now present our online algorithm, which updates SDAC parameters by online mirror descent (OMD) with unnormalized negentropy (equivalently, exponentiated gradient) over the sub-probability simplex $\Delta_H \subset \mathbb{R}^H$ with a fixed step size $\eta > 0$.

For each time t, let $M_t \in \Delta_H$ be the online parameter. We initialize at

$$M_1^{(i)} \triangleq \frac{1}{H+1}, \qquad i = 1, \ldots, H \tag{15}$$

so that the Bregman diameter of Δ_H relative to M_1 under unnormalized negentropy is at most $\log(H+1)$ (see the proof of Theorem 1).

a) Control-action selection: The controller first forms the unconstrained SDAC control action

$$\widehat{u}_t \triangleq \langle M_t, \widetilde{w}_t \rangle, \tag{16}$$

then applies a feasibility projection onto the admissible action interval $[0, \alpha x_t^{\mathcal{A}}]$:

$$u_t^{\mathcal{A}} \triangleq \min\{\widehat{u}_t,\, \alpha x_t^{\mathcal{A}}\}. \tag{17}$$

The state update is

$$x_{t+1}^{\mathcal{A}} = \alpha x_t^{\mathcal{A}} - u_t^{\mathcal{A}} + w_t, \qquad x_1^{\mathcal{A}} = 0. \tag{18}$$

b) Feasibility of the online trajectory: We verify that the algorithm satisfies (2) at every time. The key mechanism is the feasibility projection in (17), which maps the unconstrained control action onto the currently available feasible interval $[0, \alpha x_t^{\mathcal{A}}]$. Since $M_t \in \Delta_H$ and $\widetilde{w}_t \geq 0$ componentwise, equation (16) gives $\widehat{u}_t \geq 0$. By (17),

$$u_t^{\mathcal{A}} = \min\{\widehat{u}_t, \alpha x_t^{\mathcal{A}}\} \in [0, \alpha x_t^{\mathcal{A}}],$$

so the control-action feasibility constraint holds whenever $x_t^{\mathcal{A}} \geq 0$. It remains to prove $x_t^{\mathcal{A}} \geq 0$ for all t. The base case is $x_1^{\mathcal{A}} = 0$. If $x_t^{\mathcal{A}} \geq 0$, then $u_t^{\mathcal{A}} \leq \alpha x_t^{\mathcal{A}}$ and $w_t \in [0,1]$, so from (18),

$$x_{t+1}^{\mathcal{A}} = \alpha x_t^{\mathcal{A}} - u_t^{\mathcal{A}} + w_t \geq 0.$$

Thus, by induction, $x_t^{\mathcal{A}} \geq 0$ and $0 \leq u_t^{\mathcal{A}} \leq \alpha x_t^{\mathcal{A}}$ for all t.

c) Surrogate cost and OMD update: Define the surrogate cost at time t by

$$\ell_t(M) \triangleq c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big), \qquad M \in \Delta_H, \tag{19}$$

which is convex and Lipschitz by Lemma 3. Let $g_t \in \partial \ell_t(M_t)$ be a subgradient. Let

$$\Psi(M) \triangleq \sum_{i=1}^{H} \big(M^{(i)} \log M^{(i)} - M^{(i)}\big), \qquad M \in \mathbb{R}^H_{\geq 0}, \tag{20}$$

denote the unnormalized negentropy (with the convention $0 \log 0 \triangleq 0$), and let

$$D_\Psi(M \| V) \triangleq \sum_{i=1}^{H} \Big(M^{(i)} \log \tfrac{M^{(i)}}{V^{(i)}} - M^{(i)} + V^{(i)}\Big)$$

be its Bregman divergence, the generalized Kullback–Leibler divergence.

We define the OMD update

$$M_{t+1} \triangleq \operatorname*{arg\,min}_{M \in \Delta_H} \Big\{\eta \langle g_t, M \rangle + D_\Psi(M \| M_t)\Big\}. \tag{21}$$

All of the regret analysis in Section VI works directly with this optimization-based definition.

d) Closed-form solution: Although (21) is a constrained optimization problem, it is easy to solve analytically because Ψ is separable and the objective is linear in g_t. As shown in Lemma 6, the minimizer in (21) is exactly the result of two elementary steps: a multiplicative step,

$$\widetilde{M}_{t+1}^{(i)} \triangleq M_t^{(i)} \exp\big(-\eta g_t^{(i)}\big), \qquad i = 1, \ldots, H, \tag{22}$$

followed by the Bregman projection onto Δ_H, triggered only when the resulting mass exceeds one:

$$M_{t+1}^{(i)} \triangleq \frac{\widetilde{M}_{t+1}^{(i)}}{\max\big(1, S_{t+1}\big)}, \quad S_{t+1} \triangleq \sum_{j=1}^{H} \widetilde{M}_{t+1}^{(j)}, \tag{23}$$

for $i = 1, \ldots, H$. This is the form we implement in practice.

This yields a feasible online trajectory satisfying (2) by (17) and performs OMD over Δ_H through (19) and (21).

VI. Regret Analysis

The regret proof has two parts: a standard online convex optimization bound for the surrogate costs ℓ_t, and a stability bound showing that the feasibility-projected online control action stays close to the SDAC policy trajectory driven by the same parameters. The technical lemmas are deferred to Appendix A.

Theorem 1 (Regret bound)**.** *Suppose each cost function c_t is convex and satisfies* (4)*. Then the OMD controller with feasibility projection from Section V, initialized as in* (15) *and using parameter step size $\eta > 0$, satisfies*

$$\mathrm{Reg}_{\mathrm{SDAC}}(T) \leq \frac{\log(H+1)}{\eta} + \eta L^2 T \Big(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Big),$$
$$L := \alpha L_u + \tfrac{L_x}{1-\alpha}.$$

In particular, the choice

$$\eta^* = \sqrt{\frac{\log(H+1)}{L^2 T \big(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2}\big)}}$$

gives

$$\mathrm{Reg}_{\mathrm{SDAC}}(T) \leq 2L \sqrt{\Big(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Big) T \log(H+1)}.$$

Combining Theorem 1 with the approximation guarantee of Lemma 4 shows that the controller is competitive not only with the best fixed SDAC policy but with the entire family of feasible linear state-feedback policies.

Corollary 1 (No-regret against linear state-feedback policies)**.** *Run the OMD controller with feasibility projection of Section V with the step size η^* of Theorem 1. Fix any gain $q \in (0, \alpha)$, set $\beta \triangleq \alpha - q$, and let $\big(x_t^{\mathrm{lin}}, u_t^{\mathrm{lin}}\big)$ be the trajectory of the linear policy π_q^{lin} from Lemma 4. Define*

$$\mathrm{Reg}_{\mathrm{lin}}(T) \triangleq \sum_{t=1}^{T} c_t\big(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}\big) - \sum_{t=1}^{T} c_t\big(x_t^{\mathrm{lin}}, u_t^{\mathrm{lin}}\big).$$

Then

$$\mathrm{Reg}_{\mathrm{lin}}(T) \leq 2L \sqrt{\Big(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Big) T \log(H+1)}$$
$$+ \Big(L_u + \frac{L_x}{1-\alpha}\Big) \frac{q\, \beta^H}{1-\beta}\, T,$$

where $L = \alpha L_u + \frac{L_x}{1-\alpha}$. In particular, choosing the memory horizon $H = \lceil \log T / \log(1/\beta) \rceil$ forces $\beta^H \le 1/T$, so the second term is $O(1)$ and

$$\mathrm{Reg}_{\mathrm{lin}}(T) = O\Big(\sqrt{T \log\log T}\Big).$$

Thus $\mathrm{Reg}_{\mathrm{lin}}(T) = o(T)$ against every fixed linear state-feedback policy $u_t = q\, x_t$, $q \in (0,\alpha)$: the controller is no-regret relative to this class.

Proof. By the regret definition (6) and the optimized bound of Theorem 1,

$$\sum_{t=1}^{T} c_t\big(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}\big) \le \min_{M\in\Delta_H} \sum_{t=1}^{T} c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big) + 2L\sqrt{\Big(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Big) T \log(H+1)}.$$

Since $M_q^\star \in \Delta_H$ by Lemma 4, the minimum is no larger than the cost of $\pi(M_q^\star)$, which by (14) obeys

$$\sum_{t=1}^{T} c_t\big(x_t^{\pi(M_q^\star)}, u_t^{\pi(M_q^\star)}\big) \le \sum_{t=1}^{T} c_t\big(x_t^{\mathrm{lin}}, u_t^{\mathrm{lin}}\big) + \Big(L_u + \frac{L_x}{1-\alpha}\Big)\frac{q\,\beta^H}{1-\beta}\,T.$$

Chaining the two displays gives, by the definition of $\mathrm{Reg}_{\mathrm{lin}}(T)$ above, the stated bound. For the asymptotic statement, $H = \lceil \log T/\log(1/\beta)\rceil$ gives $\beta^H \le 1/T$, so the second term is at most $\big(L_u + \frac{L_x}{1-\alpha}\big)\frac{q}{1-\beta} = O(1)$, while $\log(H+1) = O(\log\log T)$ in the first term. □

VII. Conclusion

We studied adversarial online control of a storage system under state-dependent action constraints. The SDAC policy class enforces feasibility by construction, and the OMD controller with feasibility projection over SDAC parameters achieves sublinear regret $\mathrm{Reg}_{\mathrm{SDAC}}(T)$ against the best fixed SDAC policy, with only logarithmic dependence on the memory horizon. The same guarantees extend to competition with feasible linear state-feedback policies, i.e. sublinear $\mathrm{Reg}_{\mathrm{lin}}(T)$, via the SDAC approximation result. The framework is directly applicable to energy-harvesting battery management and related storage settings in which decisions must respect state constraints in real time.

Appendix A
Deferred Proofs

This appendix collects the deferred proof of the linear-state-feedback approximation guarantee from Section IV, together with the OMD and stability lemmas leading up to the regret bound stated in Section VI.

Recall that we set $w_\tau \equiv 0$ for $\tau \le 0$, which is used in several base-case arguments below.

We first prove the linear-policy approximation guarantee stated in Section IV-C.

Proof of Lemma 4. Fix $q \in (0,\alpha)$, set $\beta = \alpha - q \in (0,\alpha)$, and recall $w_\tau \equiv 0$ for $\tau \le 0$.

Feasibility and closed form of the linear policy. Substituting $u_t^{\mathrm{lin}} = q\, x_t^{\mathrm{lin}}$ into its state update gives the closed loop $x_{t+1}^{\mathrm{lin}} = \alpha x_t^{\mathrm{lin}} - q x_t^{\mathrm{lin}} + w_t = \beta x_t^{\mathrm{lin}} + w_t$ with $x_1^{\mathrm{lin}} = 0$. Since $\beta > 0$ and $w_t \ge 0$, induction yields $x_t^{\mathrm{lin}} \ge 0$, and then $0 \le u_t^{\mathrm{lin}} = q x_t^{\mathrm{lin}} \le \alpha x_t^{\mathrm{lin}}$ because $q \le \alpha$; thus π_q^{lin} satisfies (2). Unrolling the closed-loop recursion from the initialization $x_1^{\mathrm{lin}} = 0$ (1),

$$x_t^{\mathrm{lin}} = \sum_{j=1}^{t-1} \beta^{j-1} w_{t-j}, \qquad u_t^{\mathrm{lin}} = q \sum_{j=1}^{t-1} \beta^{j-1} w_{t-j}.$$

Part 1. Each $M_q^{\star(i)} = \frac{q}{\alpha}(\beta/\alpha)^{i-1} \ge 0$. Using $1 - \beta/\alpha = (\alpha-\beta)/\alpha = q/\alpha$ and the finite geometric sum,

$$\sum_{i=1}^{H} M_q^{\star(i)} = \frac{q}{\alpha}\cdot\frac{1-(\beta/\alpha)^H}{1-\beta/\alpha} = \frac{q}{\alpha}\cdot\frac{\alpha}{q}\big(1-(\beta/\alpha)^H\big) = 1 - (\beta/\alpha)^H < 1,$$

since $0 < \beta/\alpha < 1$. Hence $M_q^\star \in \Delta_H$ and $\pi(M_q^\star) \in \mathrm{SDAC}_H$.

Part 2. By the SDAC definition (9) and the identity $M_q^{\star(i)}\alpha^i = q\beta^{i-1}$,

$$u_t^{\pi(M_q^\star)} = \sum_{i=1}^{H} M_q^{\star(i)} \alpha^i w_{t-i} = q\sum_{i=1}^{H}\beta^{i-1} w_{t-i}.$$

Because $w_{t-i} = 0$ for $i \ge t$, this equals $q\sum_{i=1}^{\min\{H,t-1\}}\beta^{i-1}w_{t-i}$, so subtracting it from u_t^{lin} cancels the first $\min\{H, t-1\}$ terms:

$$u_t^{\mathrm{lin}} - u_t^{\pi(M_q^\star)} = q\sum_{j=H+1}^{t-1}\beta^{j-1}w_{t-j} \ge 0,$$

which equals 0 when $t \le H+1$. Using $w_{t-j} \in [0,1]$ and summing the geometric tail,

$$u_t^{\mathrm{lin}} - u_t^{\pi(M_q^\star)} \le q\sum_{j=H+1}^{\infty}\beta^{j-1} = \frac{q\,\beta^H}{1-\beta}.$$

Part 3. Let $\delta_t \triangleq x_t^{\pi(M_q^\star)} - x_t^{\mathrm{lin}}$ and $e_t \triangleq u_t^{\mathrm{lin}} - u_t^{\pi(M_q^\star)} \in \big[0,\, q\beta^H/(1-\beta)\big]$ by Part 2. The two trajectories follow the same dynamics (3) driven by the same disturbance, so the disturbance cancels on subtraction:

$$\delta_{t+1} = \alpha\,\delta_t - \big(u_t^{\pi(M_q^\star)} - u_t^{\mathrm{lin}}\big) = \alpha\,\delta_t + e_t, \qquad \delta_1 = 0.$$

Unrolling gives $\delta_t = \sum_{s=1}^{t-1}\alpha^{t-1-s}e_s \ge 0$, and since $e_s \le q\beta^H/(1-\beta)$ and $\sum_{s=1}^{t-1}\alpha^{t-1-s} \le 1/(1-\alpha)$,

$$0 \le \delta_t \le \frac{q\,\beta^H}{1-\beta}\cdot\frac{1}{1-\alpha} = \frac{q\,\beta^H}{(1-\beta)(1-\alpha)}.$$

Part 4. By the Lipschitz condition (4) and Parts 2–3, for each t,

$$\big|c_t\big(x_t^{\pi(M_q^\star)}, u_t^{\pi(M_q^\star)}\big) - c_t\big(x_t^{\mathrm{lin}}, u_t^{\mathrm{lin}}\big)\big| \le L_x\,|\delta_t| + L_u\,|e_t| \le \Big(\frac{L_x}{1-\alpha} + L_u\Big)\frac{q\,\beta^H}{1-\beta}.$$

Summing over $t = 1, \dots, T$ and applying the triangle inequality yields Part 4. □

The remaining lemmas establish the OMD and stability estimates used in the proof of Theorem 1; they rely on the state-decomposition, feasibility, and surrogate-cost lemmas already established in Section IV (Lemmas 1, 2, and 3).

Lemma 5 (Bregman projection onto Δ_H)**.** *Let Ψ and D_Ψ be the unnormalized negentropy and its Bregman divergence from* (20)*. Fix $\widetilde{M} \in \mathbb{R}^H_{>0}$ and set $S \triangleq \sum_{i=1}^H \widetilde{M}^{(i)}$. Then the Bregman projection of $\widetilde{M}$ onto $\Delta_H = \{M \in \mathbb{R}^H_{\geq 0} : \sum_i M^{(i)} \leq 1\}$,*

$$P(\widetilde{M}) \triangleq \operatorname*{arg\,min}_{M \in \Delta_H} D_\Psi(M \| \widetilde{M}),$$

is given coordinatewise by

$$P(\widetilde{M})^{(i)} = \frac{\widetilde{M}^{(i)}}{\max(1, S)}.$$

Proof. Since $\widetilde{M}$ is fixed, the terms $+\widetilde{M}^{(i)}$ are constant, so $P(\widetilde{M})$ minimizes

$$F(M) \triangleq \sum_{i=1}^H \Big(M^{(i)} \log \tfrac{M^{(i)}}{\widetilde{M}^{(i)}} - M^{(i)} \Big)$$

over Δ_H. The function F is strictly convex on $\mathbb{R}^H_{\geq 0}$ and Δ_H is closed and convex, so the minimizer is unique; its partial derivatives are $\partial F / \partial M^{(i)} = \log\big(M^{(i)} / \widetilde{M}^{(i)}\big)$.

Case $S \leq 1$. Here $\widetilde{M} \in \Delta_H$. Setting $\partial F/\partial M^{(i)} = 0$ gives the unconstrained minimizer $M = \widetilde{M}$ over $\mathbb{R}^H_{\geq 0}$, which is feasible; hence $P(\widetilde{M}) = \widetilde{M}$.

Case $S > 1$. Here $\widetilde{M} \notin \Delta_H$, so the constraint $\sum_i M^{(i)} \leq 1$ must be active at the optimum. As $\widetilde{M}^{(i)} > 0$, the optimal coordinates are strictly positive and the nonnegativity constraints are inactive; introducing a multiplier λ for $\sum_i M^{(i)} = 1$, KKT stationarity reads

$$\log\big(M^{(i)} / \widetilde{M}^{(i)}\big) + \lambda = 0 \quad \Longrightarrow \quad M^{(i)} = \widetilde{M}^{(i)} e^{-\lambda}.$$

The active constraint $\sum_i M^{(i)} = 1$ forces $e^{-\lambda} = 1/S$, so $M^{(i)} = \widetilde{M}^{(i)}/S$; moreover $\lambda = \log S > 0$, so dual feasibility holds. Equivalently, $P(\widetilde{M})^{(i)} = \widetilde{M}^{(i)} / \max(1, S)$ in both cases. $\square$

Lemma 6 (Closed-form solution of the OMD update)**.** *Fix $M_t \in \Delta_H$, $\eta > 0$, and $g_t \in \mathbb{R}^H$, and let $\widetilde{M}_{t+1}$ be defined coordinatewise by* (22)*, i.e. $\widetilde{M}^{(i)}_{t+1} = M^{(i)}_t e^{-\eta g^{(i)}_t}$. Then*

$$\begin{aligned} &\operatorname*{arg\,min}_{M \in \Delta_H} \Big\{ \eta \langle g_t, M \rangle + D_\Psi(M \| M_t) \Big\} \\ &\qquad = \operatorname*{arg\,min}_{M \in \Delta_H} D_\Psi(M \| \widetilde{M}_{t+1}) = P(\widetilde{M}_{t+1}), \end{aligned}$$

where $P(\cdot)$ is the Bregman projection of Lemma 5. Consequently, the OMD update (21) *coincides exactly with the closed-form OMD update* (22)–(23)*, and in particular $M_{t+1} \in \Delta_H$.*

Proof. For any $M \in \mathbb{R}^H_{\geq 0}$, expand the objective using the definitions of D_Ψ from (20) and of $\widetilde{M}_{t+1}$:

$$\begin{aligned} &\eta \langle g_t, M \rangle + D_\Psi(M \| M_t) \\ &= \sum_{i=1}^H \Big(\eta g^{(i)}_t M^{(i)} + M^{(i)} \log \tfrac{M^{(i)}}{M^{(i)}_t} - M^{(i)} + M^{(i)}_t \Big) \\ &= \sum_{i=1}^H \Big(M^{(i)} \log \tfrac{M^{(i)}}{M^{(i)}_t e^{-\eta g^{(i)}_t}} - M^{(i)} + M^{(i)}_t \Big) \\ &= \sum_{i=1}^H \Big(M^{(i)} \log \tfrac{M^{(i)}}{\widetilde{M}^{(i)}_{t+1}} - M^{(i)} + \widetilde{M}^{(i)}_{t+1} \Big) \\ &\quad + \sum_{i=1}^H \big(M^{(i)}_t - \widetilde{M}^{(i)}_{t+1} \big) \\ &= D_\Psi(M \| \widetilde{M}_{t+1}) + \sum_{i=1}^H \big(M^{(i)}_t - \widetilde{M}^{(i)}_{t+1} \big). \end{aligned}$$

The second sum does not depend on M, so minimizing the left-hand side over $M \in \Delta_H$ is equivalent to minimizing $D_\Psi(M \| \widetilde{M}_{t+1})$ over $M \in \Delta_H$, which by definition is the Bregman projection $P(\widetilde{M}_{t+1})$. Lemma 5 identifies $P(\widetilde{M}_{t+1})$ coordinatewise as $\widetilde{M}^{(i)}_{t+1} / \max(1, S_{t+1})$ with $S_{t+1} = \sum_j \widetilde{M}^{(j)}_{t+1}$, which is exactly (23). $\square$

Lemma 7 (One-step movement of OMD)**.** *For the OMD update with step size $\eta > 0$ and subgradient $g_t \in \partial \ell_t(M_t)$,*

$$\|M_{t+1} - M_t\|_1 \leq \eta \, \|g_t\|_\infty.$$

In particular, if $\|g_t\|_\infty \leq L$, then $\|M_{t+1} - M_t\|_1 \leq \eta L$.

Proof. Let Ψ and D_Ψ be as in (20). By definition (21), M_{t+1} minimizes the convex function $F(M) \triangleq \eta \langle g_t, M \rangle + D_\Psi(M \| M_t)$ over the convex set Δ_H, and F is differentiable on $\mathbb{R}^H_{>0}$ with $\nabla F(M) = \eta g_t + \nabla \Psi(M) - \nabla \Psi(M_t)$. The first-order optimality condition for constrained convex minimization therefore gives, for all $M \in \Delta_H$,

$$\Big\langle g_t + \tfrac{1}{\eta} \big(\nabla \Psi(M_{t+1}) - \nabla \Psi(M_t) \big), M - M_{t+1} \Big\rangle \geq 0.$$

Taking $M = M_t$ and rearranging yields

$$\langle g_t, M_t - M_{t+1} \rangle \geq \tfrac{1}{\eta} \Big\langle \nabla \Psi(M_{t+1}) - \nabla \Psi(M_t), \, M_{t+1} - M_t \Big\rangle.$$

By the Bregman identity,

$$\begin{aligned} &\Big\langle \nabla \Psi(M_{t+1}) - \nabla \Psi(M_t), \, M_{t+1} - M_t \Big\rangle \\ &\qquad = D_\Psi(M_{t+1} \| M_t) + D_\Psi(M_t \| M_{t+1}). \end{aligned}$$

Hence, using Hölder's inequality,

$$\begin{aligned} &D_\Psi(M_{t+1} \| M_t) + D_\Psi(M_t \| M_{t+1}) \\ &\qquad \leq \eta \, \|g_t\|_\infty \, \|M_{t+1} - M_t\|_1. \end{aligned}$$

Since both iterates lie in Δ_H, the unnormalized negentropy is 1-strongly convex with respect to $\| \cdot \|_1$ on this set: the Hessian $\nabla^2 \Psi(M) = \operatorname{diag}(1/M^{(i)})$ satisfies $v^\top \nabla^2 \Psi(M) \, v = \sum_i (v^{(i)})^2 / M^{(i)} \geq \|v\|_1^2 / (\sum_i M^{(i)}) \geq \|v\|_1^2$ whenever

$\sum_i M^{(i)} \leq 1$ (Cauchy–Schwarz), so the usual Pinsker-type bound for the Bregman divergence extends to the sub-probability simplex Δ_H,

$$D_\Psi(M_{t+1}\|M_t) \geq \tfrac{1}{2}\|M_{t+1} - M_t\|_1^2,$$
$$D_\Psi(M_t\|M_{t+1}) \geq \tfrac{1}{2}\|M_{t+1} - M_t\|_1^2,$$

and the left-hand side is at least $\|M_{t+1} - M_t\|_1^2$. Therefore

$$\|M_{t+1} - M_t\|_1^2 \leq \eta\,\|g_t\|_\infty\,\|M_{t+1} - M_t\|_1,$$

which implies $\|M_{t+1} - M_t\|_1 \leq \eta\|g_t\|_\infty$. □

Lemma 8 (Cumulative movement). *For every $t \geq 2$ and every integer i with $1 \leq i \leq t-1$,*

$$\|M_{t-i} - M_t\|_1 \leq i\,\eta L.$$

Proof. The range $1 \leq i \leq t-1$ ensures that every index $t-j$ for $j = 1, \dots, i$ satisfies $t - j \geq 1$. By the triangle inequality and Lemma 7,

$$\|M_{t-i} - M_t\|_1 \leq \sum_{j=1}^{i} \|M_{t-j} - M_{t-j+1}\|_1 \leq i\,\eta L.$$

□

Lemma 9 (Control-action bounds relative to the benchmark trajectory). *For all t,*

$$u_t^{\mathcal{A}} \geq \langle M_t, \widetilde{w}_t\rangle - \alpha\,[x_t^{\pi(M_t)} - x_t^{\mathcal{A}}]_+,$$
$$u_t^{\mathcal{A}} \leq \langle M_t, \widetilde{w}_t\rangle.$$

Proof. Recall from (16)–(17) that

$$\widehat{u}_t = \langle M_t, \widetilde{w}_t\rangle, \qquad u_t^{\mathcal{A}} = \min\{\widehat{u}_t, \alpha x_t^{\mathcal{A}}\}.$$

Therefore,

$$u_t^{\mathcal{A}} = \widehat{u}_t - \big[\widehat{u}_t - \alpha x_t^{\mathcal{A}}\big]_+ = \langle M_t, \widetilde{w}_t\rangle - \big[\langle M_t, \widetilde{w}_t\rangle - \alpha x_t^{\mathcal{A}}\big]_+.$$

This identity gives the upper bound immediately, since the subtracted term is nonnegative.

For the lower bound, Lemma 2 applied to the benchmark policy $\pi(M_t)$ yields, using (11),

$$u_t^{\pi(M_t)} = \langle M_t, \widetilde{w}_t\rangle \leq \alpha x_t^{\pi(M_t)}.$$

Subtracting $\alpha x_t^{\mathcal{A}}$ from both sides and applying monotonicity of $[\cdot]_+$ gives

$$\begin{aligned}&\big[\langle M_t, \widetilde{w}_t\rangle - \alpha x_t^{\mathcal{A}}\big]_+ \\ &\quad \leq \big[\alpha x_t^{\pi(M_t)} - \alpha x_t^{\mathcal{A}}\big]_+ = \alpha\big[x_t^{\pi(M_t)} - x_t^{\mathcal{A}}\big]_+.\end{aligned}$$

Substituting this bound into the expression above for $u_t^{\mathcal{A}}$ yields the stated lower bound. □

Lemma 10 (State-gap recursion). *Define the state gap*

$$\Gamma_t^M \triangleq x_t^{\pi(M)} - x_t^{\mathcal{A}}. \tag{24}$$

For any fixed $M \in \Delta_H$ and every $t \geq 1$ the following two-sided recursion holds:

$$\Gamma_{t+1}^M \geq \alpha\Gamma_t^M + \langle M_t - M, \widetilde{w}_t\rangle - \alpha[\Gamma_t^{M_t}]_+, \tag{25}$$
$$\Gamma_{t+1}^M \leq \alpha\Gamma_t^M + \langle M_t - M, \widetilde{w}_t\rangle. \tag{26}$$

Proof. From the state updates (12) (for $\pi(M)$) and (18) (for the algorithm), the disturbance w_t cancels and we obtain the exact identity

$$\Gamma_{t+1}^M = x_{t+1}^{\pi(M)} - x_{t+1}^{\mathcal{A}} = \alpha\Gamma_t^M - \big(u_t^{\pi(M)} - u_t^{\mathcal{A}}\big). \tag{27}$$

Consequently, bounding Γ_{t+1}^M reduces to bounding the control-action difference $u_t^{\pi(M)} - u_t^{\mathcal{A}}$.

Using the SDAC representation (11) for the benchmark control action and the bounds of Lemma 9 for the algorithm control action, we have

$$u_t^{\pi(M)} = \langle M, \widetilde{w}_t\rangle,$$

and

$$\langle M_t, \widetilde{w}_t\rangle - \alpha[\Gamma_t^{M_t}]_+ \leq u_t^{\mathcal{A}} \leq \langle M_t, \widetilde{w}_t\rangle.$$

Subtracting the upper bound on $u_t^{\mathcal{A}}$ from $u_t^{\pi(M)}$ yields the lower bound on $u_t^{\pi(M)} - u_t^{\mathcal{A}}$:

$$u_t^{\pi(M)} - u_t^{\mathcal{A}} \geq \langle M, \widetilde{w}_t\rangle - \langle M_t, \widetilde{w}_t\rangle = -\langle M_t - M, \widetilde{w}_t\rangle.$$

Inserting this into the identity (27) gives

$$\Gamma_{t+1}^M \leq \alpha\Gamma_t^M + \langle M_t - M, \widetilde{w}_t\rangle,$$

which is (26).

Similarly, using the lower bound on $u_t^{\mathcal{A}}$ produces an upper bound on $u_t^{\pi(M)} - u_t^{\mathcal{A}}$:

$$\begin{aligned}u_t^{\pi(M)} - u_t^{\mathcal{A}} &\leq \langle M, \widetilde{w}_t\rangle - \Big(\langle M_t, \widetilde{w}_t\rangle - \alpha[\Gamma_t^{M_t}]_+\Big) \\ &= -\langle M_t - M, \widetilde{w}_t\rangle + \alpha[\Gamma_t^{M_t}]_+.\end{aligned}$$

Substituting this bound into the identity (27) yields

$$\Gamma_{t+1}^M \geq \alpha\Gamma_t^M + \langle M_t - M, \widetilde{w}_t\rangle - \alpha[\Gamma_t^{M_t}]_+,$$

which is (25) and completes the proof. □

Lemma 11 (State deviation bound). *For every $t \geq 1$, let the state gap be defined by* (24)*. Then*

$$-\frac{\alpha\,\eta L}{(1-\alpha)^3} \leq \Gamma_t^{M_t} = x_t^{\pi(M_t)} - x_t^{\mathcal{A}} \leq \frac{\alpha\,\eta L}{(1-\alpha)^2}.$$

Proof. From (26), for any fixed benchmark $M \in \Delta_H$ and every $s \geq 1$,

$$\Gamma_{s+1}^M \leq \alpha\Gamma_s^M + \langle M_s - M, \widetilde{w}_s\rangle.$$

Applying the recursion repeatedly up to time t yields

$$\begin{aligned}\Gamma_t^M &\leq \alpha\Gamma_{t-1}^M + \langle M_{t-1} - M, \widetilde{w}_{t-1}\rangle \\ &\leq \alpha^2\Gamma_{t-2}^M + \alpha\langle M_{t-2} - M, \widetilde{w}_{t-2}\rangle \\ &\qquad + \langle M_{t-1} - M, \widetilde{w}_{t-1}\rangle \\ &\ \ \vdots \\ &\leq \alpha^{t-1}\Gamma_1^M + \sum_{s=1}^{t-1}\alpha^{t-1-s}\langle M_s - M, \widetilde{w}_s\rangle.\end{aligned}$$

Now set the benchmark equal to the online parameter $M = M_t$. Since $x_1^{\pi(M_t)} = x_1^{\mathcal{A}} = 0$, we have $\Gamma_1^{M_t} = 0$, and therefore

$$\Gamma_t^{M_t} \leq \sum_{s=1}^{t-1}\alpha^{t-1-s}\langle M_s - M_t, \widetilde{w}_s\rangle.$$

Note that $\|\widetilde{w}_s\|_\infty = \max_{i\le H} \alpha^i w_{s-i} \le \alpha$ since $w_\tau \in [0,1]$ and $i \ge 1$. Hence,

$$\begin{aligned}\Gamma_t^{M_t} &\le \sum_{s=1}^{t-1} \alpha^{t-1-s} \big|\langle M_s - M_t, \widetilde{w}_s\rangle\big| \\ &\le \sum_{s=1}^{t-1} \alpha^{t-1-s} \|M_s - M_t\|_1 \|\widetilde{w}_s\|_\infty \\ &\le \alpha\,\eta L \sum_{s=1}^{t-1} \alpha^{t-1-s}(t-s) \\ &= \alpha\,\eta L \sum_{j=1}^{t-1} j\,\alpha^{j-1} \le \alpha\,\eta L \sum_{j=1}^{\infty} j\,\alpha^{j-1} = \frac{\alpha\,\eta L}{(1-\alpha)^2}.\end{aligned}$$

Where we used the geometric-series identity

$$\sum_{j=1}^{\infty} j\,\alpha^{j-1} = \frac{1}{(1-\alpha)^2}.$$

Similarly, for the lower bound, unrolling (25) with the same fixed benchmark $M = M_t$ yields

$$\Gamma_t^{M_t} \ge \sum_{s=1}^{t-1} \alpha^{t-1-s} \langle M_s - M_t, \widetilde{w}_s\rangle - \sum_{s=1}^{t-1} \alpha^{t-s} [\Gamma_s^{M_s}]_+.$$

The upper bound just proved applies to every earlier time s, so $[\Gamma_s^{M_s}]_+ \le \alpha\eta L/(1-\alpha)^2$. Also, by Hölder's inequality and Lemma 8,

$$\begin{aligned}\Gamma_t^{M_t} &\ge -\alpha\,\eta L \sum_{s=1}^{t-1} \alpha^{t-1-s}(t-s) \\ &\quad - \frac{\alpha\,\eta L}{(1-\alpha)^2} \sum_{s=1}^{t-1} \alpha^{t-s} \\ &= -\alpha\,\eta L \sum_{j=1}^{t-1} j\,\alpha^{j-1} - \frac{\alpha\,\eta L}{(1-\alpha)^2} \sum_{j=1}^{t-1} \alpha^{j}.\end{aligned}$$

Extending both sums to infinity gives

$$\Gamma_t^{M_t} \ge -\alpha\,\eta L \sum_{j=1}^{\infty} j\,\alpha^{j-1} - \frac{\alpha\,\eta L}{(1-\alpha)^2} \sum_{j=1}^{\infty} \alpha^{j}.$$

Using the geometric-series identities

$$\sum_{j=1}^{\infty} j\,\alpha^{j-1} = \frac{1}{(1-\alpha)^2}, \qquad \sum_{j=1}^{\infty} \alpha^{j} = \frac{\alpha}{1-\alpha},$$

we obtain

$$\Gamma_t^{M_t} \ge -\frac{\alpha\,\eta L}{(1-\alpha)^2} - \frac{\alpha^2\,\eta L}{(1-\alpha)^3} = -\frac{\alpha\,\eta L}{(1-\alpha)^3}.$$

Hence, the claim. □

Lemma 12 (Action-state gap)**.** *Let the state gap be defined by* (24)*. Then for all t,*

$$0 \le u_t^{\pi(M_t)} - u_t^{\mathcal{A}} \le \frac{\alpha^2\,\eta L}{(1-\alpha)^2}.$$

Proof. By (11) and (16), the benchmark control action equals the unconstrained control action, $u_t^{\pi(M_t)} = \langle M_t, \widetilde{w}_t\rangle = \widehat{u}_t$, while $u_t^{\mathcal{A}} = \min\{\widehat{u}_t, \alpha x_t^{\mathcal{A}}\} \le \widehat{u}_t$ by (17). Hence the gap is nonnegative and

$$u_t^{\pi(M_t)} - u_t^{\mathcal{A}} = \widehat{u}_t - \min\{\widehat{u}_t, \alpha x_t^{\mathcal{A}}\} = \big[\langle M_t, \widetilde{w}_t\rangle - \alpha x_t^{\mathcal{A}}\big]_+.$$

As established in the proof of Lemma 9 (using feasibility of $\pi(M_t)$),

$$\big[\langle M_t, \widetilde{w}_t\rangle - \alpha x_t^{\mathcal{A}}\big]_+ \le \alpha\,[x_t^{\pi(M_t)} - x_t^{\mathcal{A}}]_+ = \alpha\,[\Gamma_t^{M_t}]_+.$$

By Lemma 11, $[\Gamma_t^{M_t}]_+ \le \dfrac{\alpha\,\eta L}{(1-\alpha)^2}$, so

$$u_t^{\pi(M_t)} - u_t^{\mathcal{A}} \le \alpha\,[\Gamma_t^{M_t}]_+ \le \frac{\alpha^2\,\eta L}{(1-\alpha)^2},$$

which completes the proof. □

Lemma 13 (Stability term)**.** *For all horizons T,*

$$\sum_{t=1}^{T} \Big(c_t(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big) \Big) \le \frac{\alpha\,T\eta L^2}{(1-\alpha)^2},$$

where $L = \alpha L_u + \frac{L_x}{1-\alpha}$.

Proof. By Lipschitz continuity of c_t (see equation (4)),

$$\begin{aligned}&c_t\big(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}\big) - c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big) \\ &\qquad \le L_u\big|u_t^{\mathcal{A}} - u_t^{\pi(M_t)}\big| + L_x\big|x_t^{\mathcal{A}} - x_t^{\pi(M_t)}\big|.\end{aligned}$$

From Lemma 12,

$$|u_t^{\mathcal{A}} - u_t^{\pi(M_t)}| \le \frac{\alpha^2\,\eta L}{(1-\alpha)^2},$$

Also, by Lemma 11 and (24),

$$|x_t^{\mathcal{A}} - x_t^{\pi(M_t)}| = |\Gamma_t^{M_t}| \le \frac{\alpha\,\eta L}{(1-\alpha)^3}.$$

Therefore, for each $t \ge 1$,

$$\begin{aligned}&c_t\big(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}\big) - c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big) \\ &\qquad \le L_u \frac{\alpha^2\,\eta L}{(1-\alpha)^2} + L_x \frac{\alpha\,\eta L}{(1-\alpha)^3}.\end{aligned}$$

Summing over $t = 1, 2, \ldots, T$ gives

$$\begin{aligned}&\sum_{t=1}^{T} \Big(c_t(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - c_t(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}) \Big) \\ &\qquad \le \frac{\alpha\,T\eta L}{(1-\alpha)^2} \left(\alpha L_u + \frac{L_x}{1-\alpha}\right) = \frac{\alpha\,T\eta L^2}{(1-\alpha)^2},\end{aligned}$$

where we used $L = \alpha L_u + \frac{L_x}{1-\alpha}$ in the final step. □

Proof of Theorem 1. Starting from (6) we have the following decomposition of the regret:

$$\begin{aligned}\mathrm{Reg}_{\text{SDAC}}(T) &= \sum_{t=1}^{T} c_t(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}) \\ &\quad - \min_{M\in\Delta_H}\sum_{t=1}^{T} c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big) \\ &= \sum_{t=1}^{T} c_t(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - \sum_{t=1}^{T} c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big) \\ &\quad + \sum_{t=1}^{T} c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big) \\ &\quad - \min_{M\in\Delta_H}\sum_{t=1}^{T} c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big).\end{aligned}$$

We call the first sum the stability term, and the second sum the OMD term. We will bound each term separately.

We bound the OMD term using the OMD update (21) with the unnormalized negentropy Ψ and Bregman divergence D_Ψ from (20). Under the initialization (15), for any benchmark $M^\star \in \Delta_H$ with $W^\star \triangleq \sum_{i=1}^{H} M^{\star(i)} \le 1$,

$$\begin{aligned}&D_\Psi(M^\star \| M_1) \\ &\quad = \sum_{i=1}^{H}\Big(M^{\star(i)}\log\big((H+1)M^{\star(i)}\big) - M^{\star(i)} + \tfrac{1}{H+1}\Big) \\ &\quad = \sum_{i=1}^{H} M^{\star(i)}\log M^{\star(i)} + W^\star\log(H+1) - W^\star + \tfrac{H}{H+1}.\end{aligned}$$

Since each $M^{\star(i)} \in [0,1]$, one has $M^{\star(i)}\log M^{\star(i)} \le 0$, and therefore

$$D_\Psi(M^\star\|M_1) \le W^\star\log(H+1) - W^\star + \tfrac{H}{H+1}.$$

The right-hand side is at most $\log(H+1)$ for every $W^\star \in [0,1]$: equivalently,

$$(1-W^\star)\log(H+1) + W^\star - \tfrac{H}{H+1} \ge 0,$$

which holds because the left-hand side is affine in $W^\star$ and nonnegative at the endpoints $W^\star = 0$ and $W^\star = 1$. Hence $D_\Psi(M^\star\|M_1) \le \log(H+1)$. The surrogate costs ℓ_t from (19) are convex and L-Lipschitz by Lemma 3, so the standard OMD regret bound with Bregman divergence D_Ψ on Δ_H [19] yields

$$\sum_{t=1}^{T}\ell_t(M_t) - \min_{M\in\Delta_H}\sum_{t=1}^{T}\ell_t(M) \le \frac{\log(H+1)}{\eta} + \frac{\eta L^2 T}{2}.$$

Using (19), we substitute $\ell_t(M) = c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big)$, so that

$$\begin{aligned}\sum_{t=1}^{T}\ell_t(M_t) &= \sum_{t=1}^{T} c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big), \\ \min_{M\in\Delta_H}\sum_{t=1}^{T}\ell_t(M) &= \min_{M\in\Delta_H}\sum_{t=1}^{T} c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big).\end{aligned}$$

Thus this term becomes

$$\begin{aligned}&\sum_{t=1}^{T} c_t\big(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}\big) \\ &\quad - \min_{M\in\Delta_H}\sum_{t=1}^{T} c_t\big(x_t^{\pi(M)}, u_t^{\pi(M)}\big) \\ &\quad \le \frac{\log(H+1)}{\eta} + \frac{\eta L^2 T}{2}.\end{aligned}$$

From Lemma 13, the stability term satisfies

$$\begin{aligned}&\sum_{t=1}^{T} c_t(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - \sum_{t=1}^{T} c_t(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}) \\ &\quad \le \frac{\alpha\,\eta L^2 T}{(1-\alpha)^2}.\end{aligned}$$

Adding the OMD term and the stability term yields the combined bound

$$\mathrm{Reg}_{\text{SDAC}}(T) \;\le\; \frac{\log(H+1)}{\eta} \;+\; \frac{\eta L^2 T}{2} \;+\; \frac{\alpha\,\eta L^2 T}{(1-\alpha)^2}.$$

Rearranging by factoring out the η term gives

$$\mathrm{Reg}_{\text{SDAC}}(T) \;\le\; \frac{\log(H+1)}{\eta} \;+\; \eta L^2 T\Big(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Big).$$

To optimize over $\eta > 0$, we apply the inequality $A/\eta + B\eta \le 2\sqrt{AB}$ with $A = \log(H+1)$ and $B = L^2T(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2})$. This yields the optimal step size

$$\eta^* = \sqrt{\frac{\log(H+1)}{L^2T\big(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2}\big)}}$$

and the resulting regret bound

$$\mathrm{Reg}_{\text{SDAC}}(T) \;\le\; 2L\sqrt{\Big(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Big)T\log(H+1)}.$$

□

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